This is a reduction of 37% – 28% = 9%. In short, by splitting your money into two parts, you are reducing considerably the risk of underperforming the CD. Thus, as you can see, even though the correlation between the two investments is zero—that is, if one does better than average, there is no indication of how the other will perform—there are still benefits to diversifying.

Similarly, if the correlation coefficient is –25%, the odds of your portfolio doing worse than the term deposit is 24% (or about one in four). Compare that with the 28% chance of doing worse when the correlation is zero. Once again, the benefits are clear. The lower the correlation between the two investments, the lower your risk.

I like to call this the fundamental law of diversification: "The risk of the sum is less than the sum of the risk."

What do I mean? Remember that you need two ingredients or factors for successful diversification. The first is nonperfect correlation; the second is an expectation of some profit from both investments. How much you benefit from diversification will depend on the strength of these two factors.

By investing in two imperfectly correlated assets, you are reducing the overall shortfall risk compared to the individual shortfall risks of each asset. Thus, the risk of the sum, which is the risk of the portfolio that consists of two assets, is less than the sum of the risks; that is, just adding the risk of each individual asset.

Let me clarify this point by walking through a detailed example. We know that if you put $100 in a term deposit that pays 5% per annum, you'll have $105 at the end of the year. Not much of a payback, but very safe. Now, let's pretend that you want to invest in the market, take some risk, and diversify. Instead of putting your money in the bank, you put $50 in one asset (let's call it Fund XYZ), and $50 in another (Fund ABC).

Now, we know from Table 3.1 that if there is zero correlation between the price movements of these two investments, then you stand a 28% chance of having less than $105 at the end of the year. This is what I call the risk of the sum. It's the risk of your capital, or the risk of your portfolio.